

Bengkel Pendedahan
Olimpiad Matematik MRSM
Last Digits & Divisibility Rules

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1 Last Digits

1.1 Strategy

To find last digits, we only need to consider the last digits when performing operations. In many cases, finding patterns is useful.

1.2 Problems

1. Find the last digit of 20^{2025} .
2. Find the last digit of 9^{2025} .
3. Find the last digit of $(1 + 5^{2025})^{2025}$.
4. (PML) Find the last digit of $2024^{2023^{2022}}$.
5. Find the last digit of $21^{21} \times 22^{22} \times 23^{23} \times 24^{24} \times 25^{25}$.
6. Find the last digit of 7^{2021} , 7^{2022} , 7^{2023} and 7^{2024} .
7. Find the last digit of $2022^{2023} + 2023^{2022}$.
8. Find the last digit of $2025!$.
9. (PML) Find the last 2 digits of $20! \times 24!$.
10. Find the last digit of $1 \times 3 \times 5 \times \cdots \times 999$.
11. Find the last 3 digits of $2 \times 4 \times 6 \times \cdots \times 2022 \times 2024$.
12. Find the last digit of $1^{2025} + 2^{2025} + 3^{2025} + \cdots + 99^{2025} + 100^{2025}$.
13. (AMC 10) Find the last 2 digits of $2015^{2016} - 2017$.

2 Divisibility Rules

2.1 Rules

- Divisible by $10^n \iff$ Last n digits are 0
- Divisible by $2^n \iff$ Last n digits divisible by 2^n
- Divisible by $5^n \iff$ Last n digits divisible by 5^n
- Divisible by 3 $\iff$ Sum of digits divisible by 3
- Divisible by 9 $\iff$ Sum of digits divisible by 9
- Divisible by 11 $\iff$ Alternating sum of digits divisible by 11

2.2 Strategy

Exploit divisibility rules to obtain information about the digits of a number.

2.3 Problems

1. (IMONST 1 2023) The number -6 is divisible by 3.
2. (IMONST 1 2023) 0 is divisible by -3 .
3. (IMONST 1 2022) Determine the number $a \times b$ if the number $\overline{a2022b}$ is divisible by 72.
4. (IMONST 1 2021) What is the smallest positive multiple of 24 that can be written using digits 4 and 5 only?
5. (IMONST 1 2020) What is the smallest positive multiple of 225 that can be written using digits 0 and 1 only?
6. (CJR) If the 3-digit number $\overline{2a7}$ is divisible by 11, find the value of a .
7. (CJR) Given that the 5-digit number $\overline{2a5a6}$ is divisible by 36, find the digit a .
8. (SMO) What is the smallest 5-digit integer of the form $\overline{5x20y}$ that is divisible by 33?
9. (AMC 12) The six-digit number $\overline{20210A}$ is prime for only one digit A . What is A ?